Project Proposal: Mitigating Temporal Blindness in Egocentric VideoQA for the Blind

Aimine Meddeb
MVA ENS Paris-Saclay
aimine28@gmail.com

November 4, 2026

1 Introduction and Motivation

Current Multimodal Large Language Models (MLLMs) struggle with the Egocentric Video Question Answering (VideoQA) task. On the recently introduced EgoBlind dataset [1] state-of-the-art models like GPT-4o and InternVL2.5 reach only $\sim$60% accuracy, far below the human performance of 87.4%.

This project addresses two important issues of current MLLMs in this domain.

- Temporal Blindness: Models often fail to sample the most relevant frames preceding the question timestamp, leading to poor spatial orientation and task understanding.
- Sycophancy: When a BLV user asks a question based on a false visual premise (for instance, the example (d) in figure 4 of [1] asking how to navigate to a bridge they are already on), models can hallucinate pleasant but dangerous answers rather than correcting the user.

2 Proposed Work and Baseline

2.1 Dataset and Code

- Dataset: EgoBlind [1], containing 1,329 egocentric videos and 5,311 timestamped questions across categories like Navigation, Tool Use, and Safety Warnings.
- Code : The official code and benchmarks are available at `https://github.com/doc-doc/EgoBlind`.

2.2 Baseline Reproduction

The first step involves reproducing the zeroshot results on the EgoBlind test set using an open-source model (for example Qwen2.5-VL-7B). This requires setting up the GPT-4o mini-based evaluation pipeline defined in the paper. The baseline relies on uniform frame sampling up to the question time $t \leq t_Q$.

3 Extension

One issue about the paper algorithm is that frames are uniformly sampled without any preference from the question asked. In order to reduce the number of frames sampled and to increase the relevance of those, this extension propose to implement a key-frame retrieval module prior to VLM inference. Instead of uniform sampling, this module will select frames based on semantic relevance to the user's question. This selection will ensure that critical frames are presented to the VLM.

3.1 Implementation:

- Feature extraction: Use a pre-trained vision-language encoder (like CLIP) to extract embeddings from the user's question and all video frames prior to t_Q.

- Adaptive sampling: Calculate similarity to select the top-K most relevant frames. In order to preserve recent context, a fixed number of frames before t_Q will also be included.
- Anti-sycophancy prompting (to be precised...): Introduce a two-stage prompting mechanism where the VLM first verifies the visual information related to the question before providing an answer. This will allow the VLM to know if it has enough information to answer in order to limit hallucinations.

4 Evaluation and Difficulties

- Evaluation: The method will be evaluated using the same metrics as EgoBlind (accuracy across 6 categories). One can expect to observe improvements particularly in the 'navigation' and 'tool use' categories.
- Difficulties: The project focuses on algorithmic design (optimizing the algorithm) rather than model fine tuning. One can estimate that the most exigent computational resource is a GPU (16GB) for running the CLIP encoder and the VLM in inference mode.

References

[1] J. Xiao et al. Egoblind: Towards egocentric visual assistance for the blind. *NeurIPS Datasets and Benchmarks*, 2025.